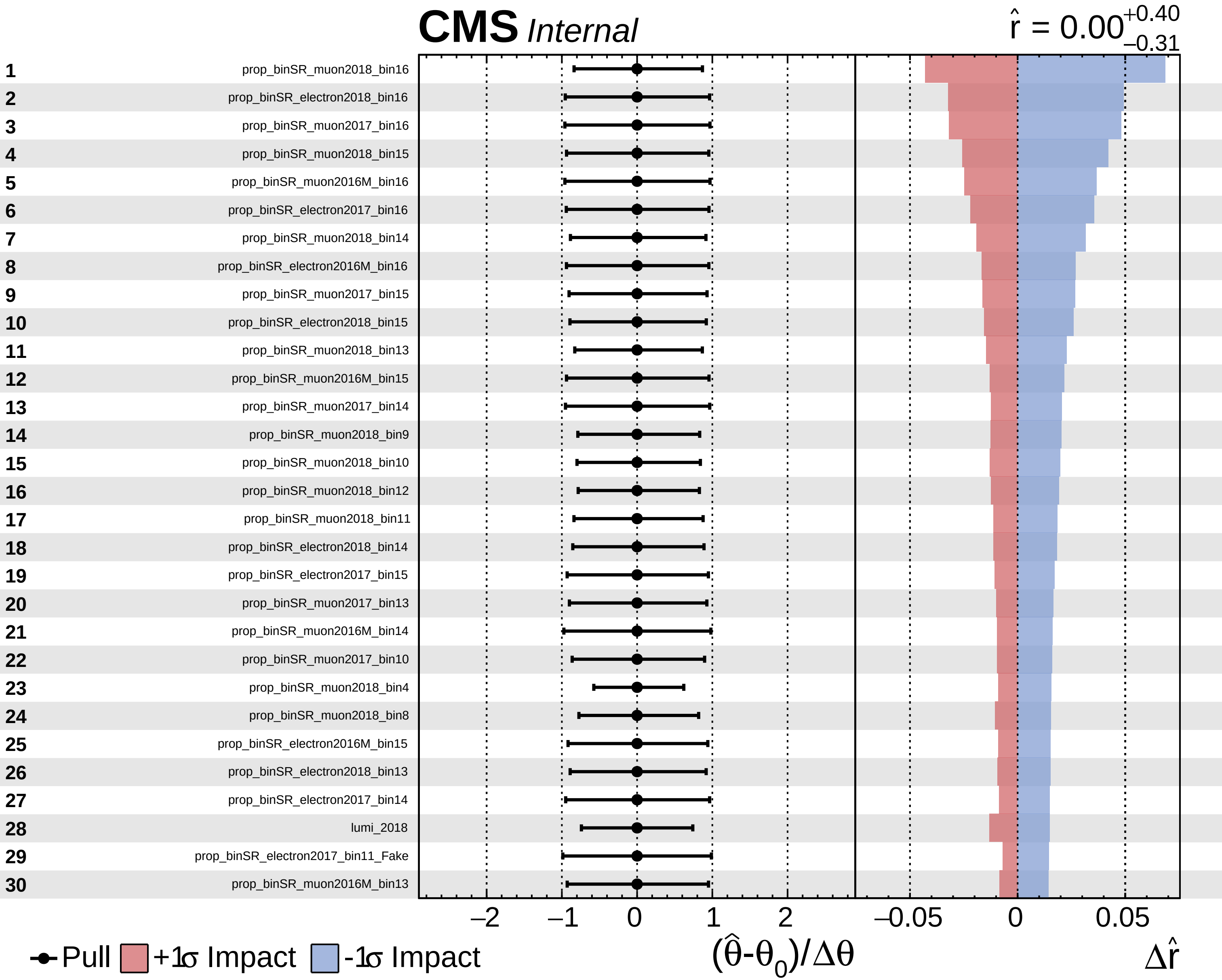
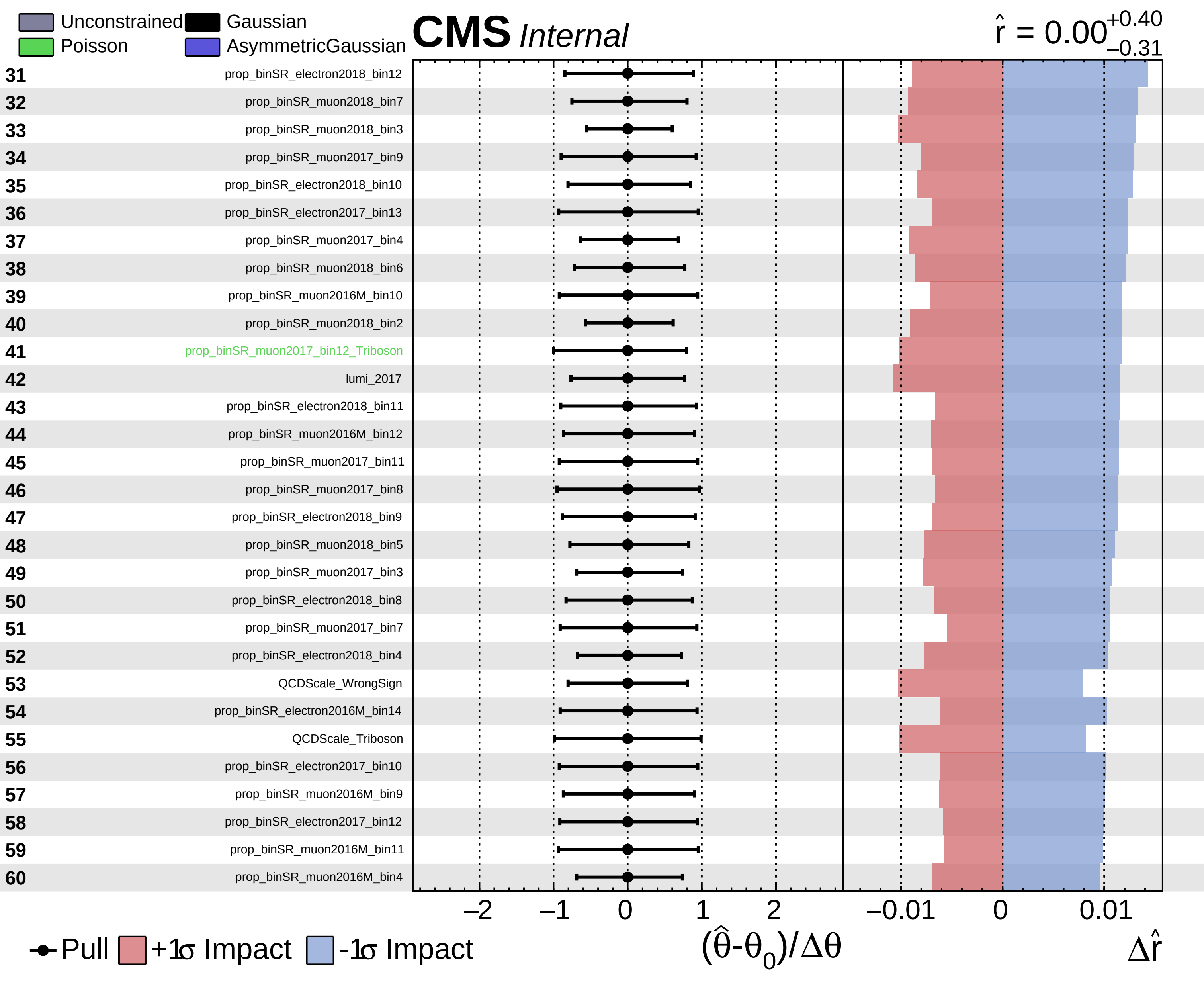
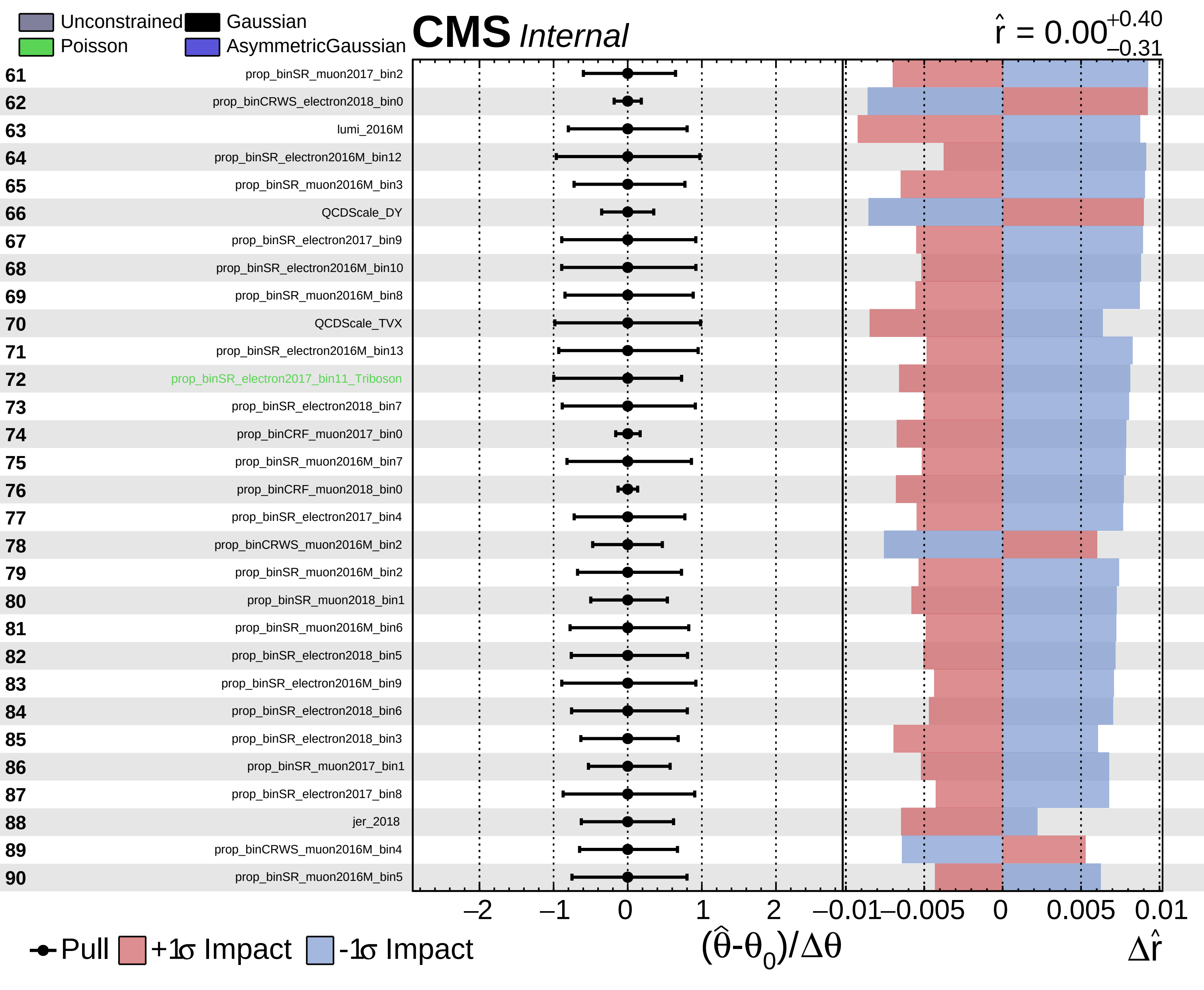
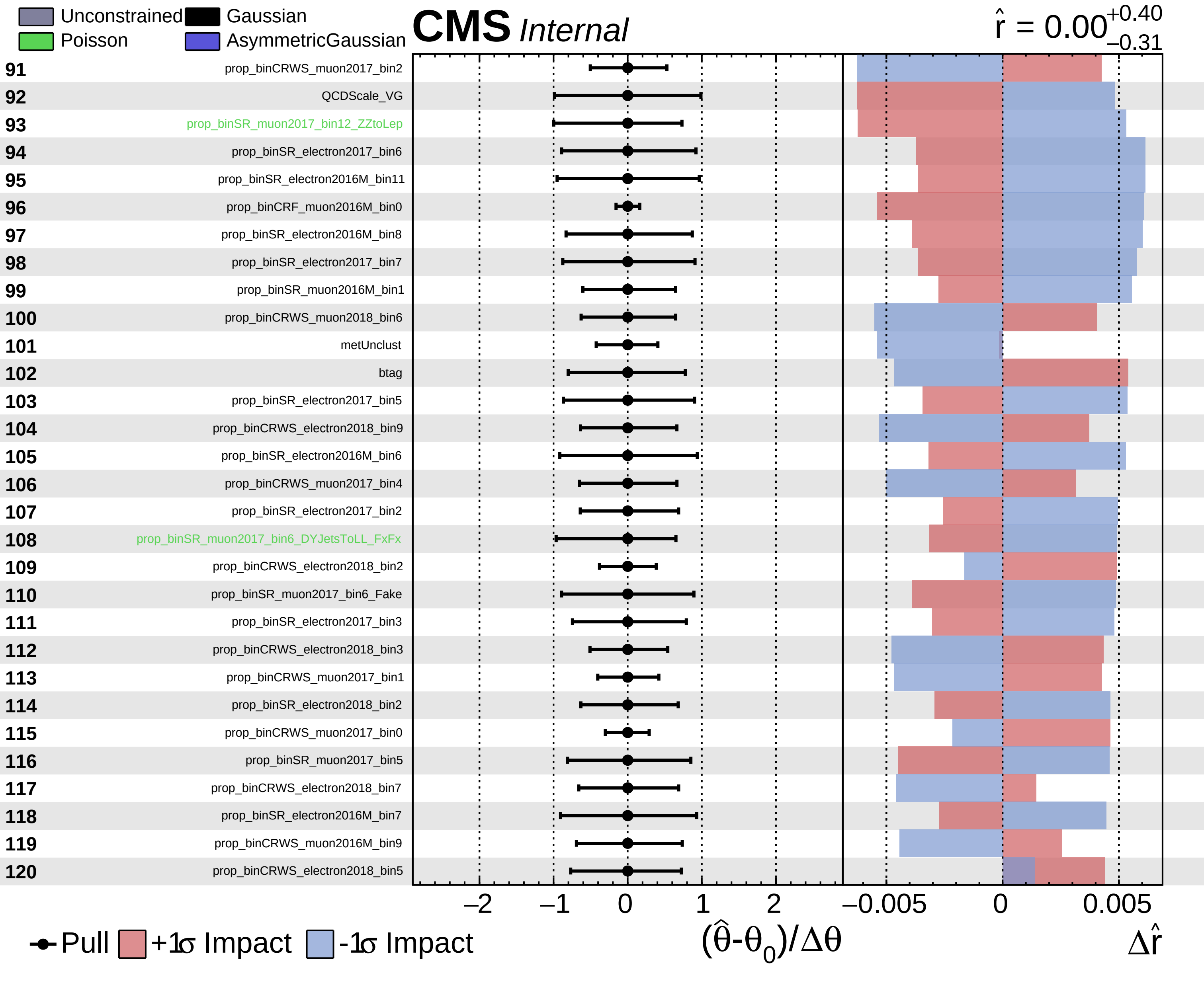


CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 prop_binSR_electron2016M_bin16
9 prop_binSR_muon2017_bin15
10 prop_binSR_electron2018_bin15
11 prop_binSR_muon2018_bin13
12 prop_binSR_muon2016M_bin15
13 prop_binSR_muon2017_bin14
14 prop_binSR_muon2018_bin9
15 prop_binSR_muon2018_bin10
16 prop_binSR_muon2018_bin12
17 prop_binSR_muon2018_bin11
18 prop_binSR_electron2018_bin14
19 prop_binSR_electron2017_bin15
20 prop_binSR_muon2017_bin13
21 prop_binSR_muon2016M_bin14
22 prop_binSR_muon2017_bin10
23 prop_binSR_muon2018_bin4
24 prop_binSR_muon2018_bin8
25 prop_binSR_electron2016M_bin15
26 prop_binSR_electron2018_bin13
27 prop_binSR_electron2017_bin14
28 lumi_2018
29 prop_binSR_electron2017_bin11_Fake
30 prop_binSR_muon2016M_bin13
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

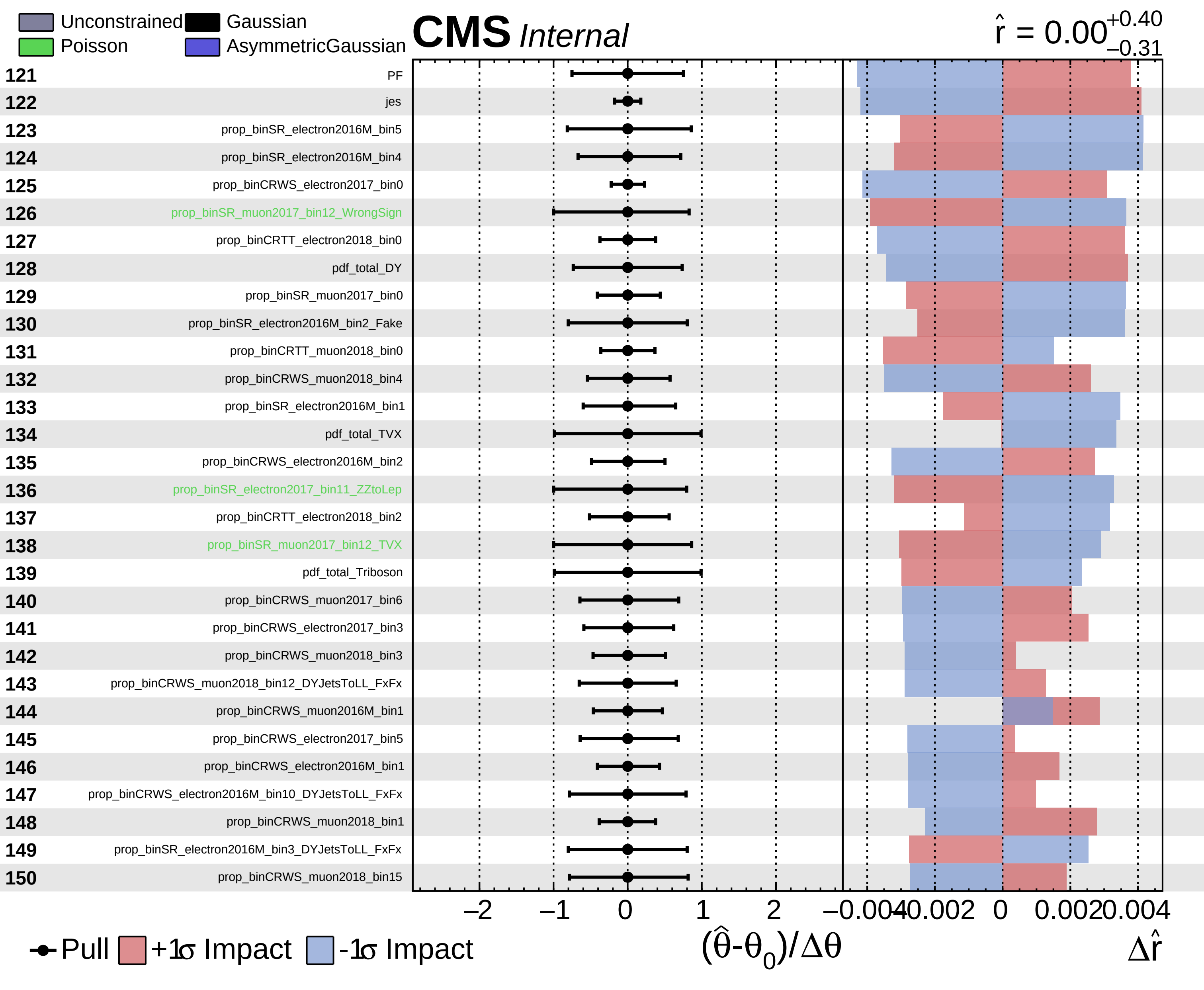

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
31 prop_binSR_electron2018_bin12
32 prop_binSR_muon2018_bin7
33 prop_binSR_muon2018_bin3
34 prop_binSR_muon2017_bin9
35 prop_binSR_electron2018_bin10
36 prop_binSR_electron2017_bin13
37 prop_binSR_muon2017_bin4
38 prop_binSR_muon2018_bin6
39 prop_binSR_muon2016M_bin10
40 prop_binSR_muon2018_bin2
41 prop_binSR_muon2017_bin12_Triboson
42 lumi_2017
43 prop_binSR_electron2018_bin11
44 prop_binSR_muon2016M_bin12
45 prop_binSR_muon2017_bin11
46 prop_binSR_muon2017_bin8
47 prop_binSR_electron2018_bin9
48 prop_binSR_muon2018_bin5
49 prop_binSR_muon2017_bin3
50 prop_binSR_electron2018_bin8
51 prop_binSR_muon2017_bin7
52 prop_binSR_electron2018_bin4
53 QCDScale_WrongSign
54 prop_binSR_electron2016M_bin14
55 QCDScale_Triboson
56 prop_binSR_electron2017_bin10
57 prop_binSR_muon2016M_bin9
58 prop_binSR_electron2017_bin12
59 prop_binSR_muon2016M_bin11
60 prop_binSR_muon2016M_bin4
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



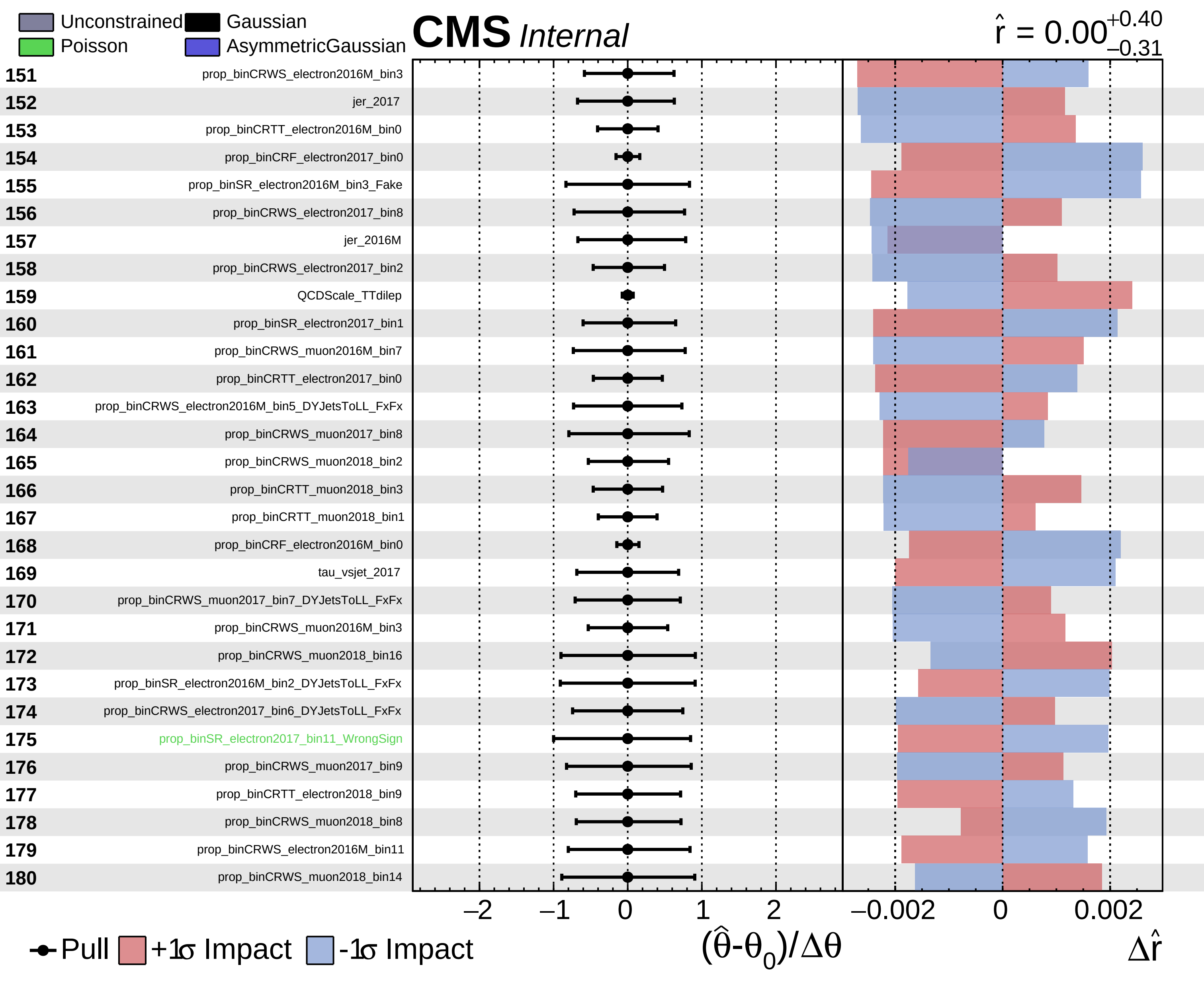
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
61 prop_binSR_muon2017_bin2
62 prop_binCRWS_electron2018_bin0
63 lumi_2016M
64 prop_binSR_electron2016M_bin12
65 prop_binSR_muon2016M_bin3
66 QCDScale_DY
67 prop_binSR_electron2017_bin9
68 prop_binSR_electron2016M_bin10
69 prop_binSR_muon2016M_bin8
70 QCDScale_TVX
71 prop_binSR_electron2016M_bin13
72 prop_binSR_electron2017_bin11_Triboson
73 prop_binSR_electron2018_bin7
74 prop_binCRF_muon2017_bin0
75 prop_binSR_muon2016M_bin7
76 prop_binCRF_muon2018_bin0
77 prop_binSR_electron2017_bin4
78 prop_binCRWS_muon2016M_bin2
79 prop_binSR_muon2016M_bin2
80 prop_binSR_muon2018_bin1
81 prop_binSR_muon2016M_bin6
82 prop_binSR_electron2018_bin5
83 prop_binSR_electron2016M_bin9
84 prop_binSR_electron2018_bin6
85 prop_binSR_electron2018_bin3
86 prop_binSR_muon2017_bin1
87 prop_binSR_electron2017_bin8
88 jer_2018
89 prop_binCRWS_muon2016M_bin4
90 prop_binSR_muon2016M_bin5
−2 −1 0 1 2
−0.01 −0.005 0 0.005 0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



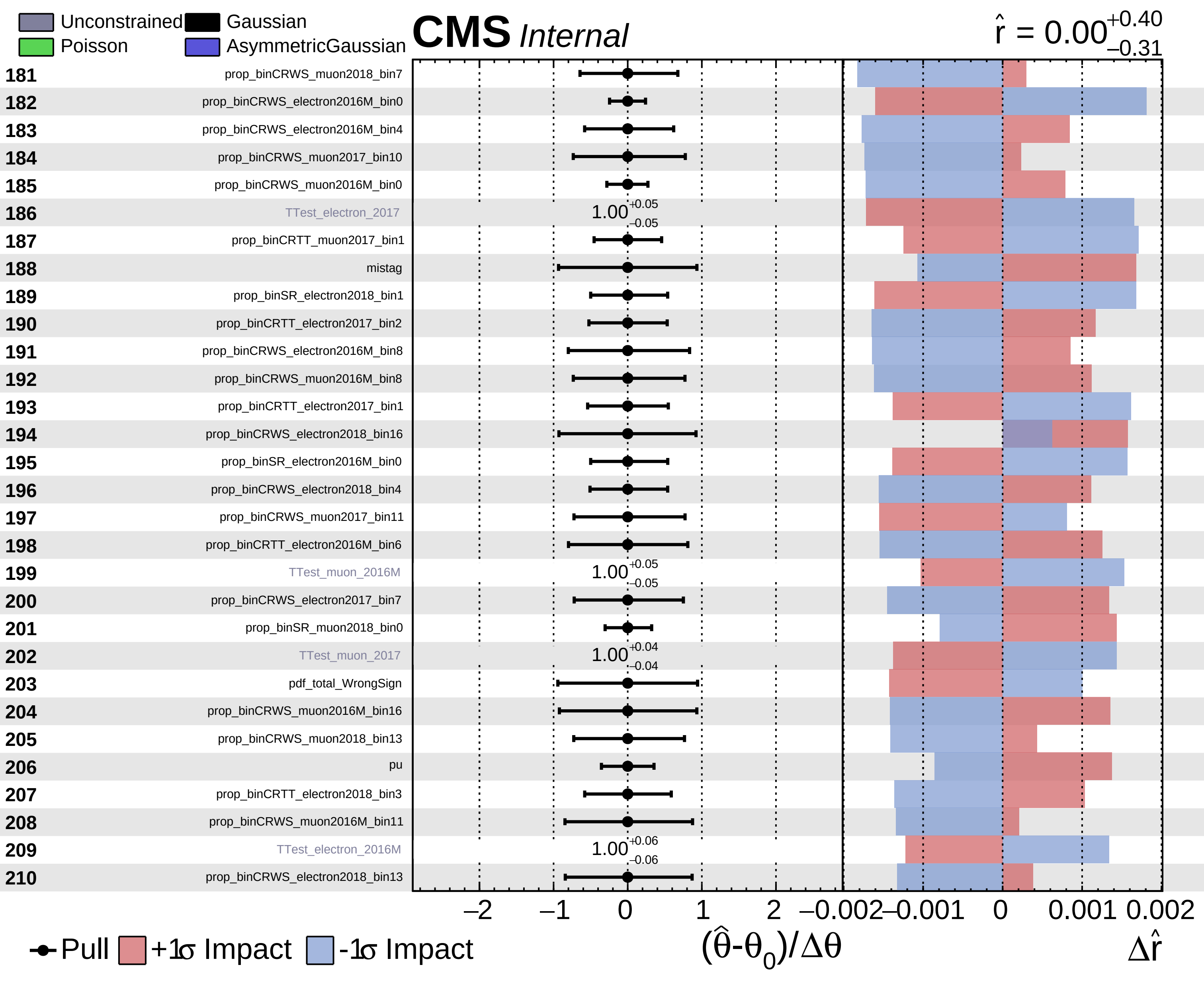
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
91 prop_binCRWS_muon2017_bin2
92 QCDScale_VG
93 prop_binSR_muon2017_bin12_ZZtoLep
94 prop_binSR_electron2017_bin6
95 prop_binSR_electron2016M_bin11
96 prop_binCRF_muon2016M_bin0
97 prop_binSR_electron2016M_bin8
98 prop_binSR_electron2017_bin7
99 prop_binSR_muon2016M_bin1
100 prop_binCRWS_muon2018_bin6
101 metUnclust
102 btag
103 prop_binSR_electron2017_bin5
104 prop_binCRWS_electron2018_bin9
105 prop_binSR_electron2016M_bin6
106 prop_binCRWS_muon2017_bin4
107 prop_binSR_electron2017_bin2
108 prop_binSR_muon2017_bin6_DYJetsToLL_FxFx
109 prop_binCRWS_electron2018_bin2
110 prop_binSR_muon2017_bin6_Fake
111 prop_binSR_electron2017_bin3
112 prop_binCRWS_electron2018_bin3
113 prop_binCRWS_muon2017_bin1
114 prop_binSR_electron2018_bin2
115 prop_binCRWS_muon2017_bin0
116 prop_binSR_muon2017_bin5
117 prop_binCRWS_electron2018_bin7
118 prop_binSR_electron2016M_bin7
119 prop_binCRWS_muon2016M_bin9
120 prop_binCRWS_electron2018_bin5
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



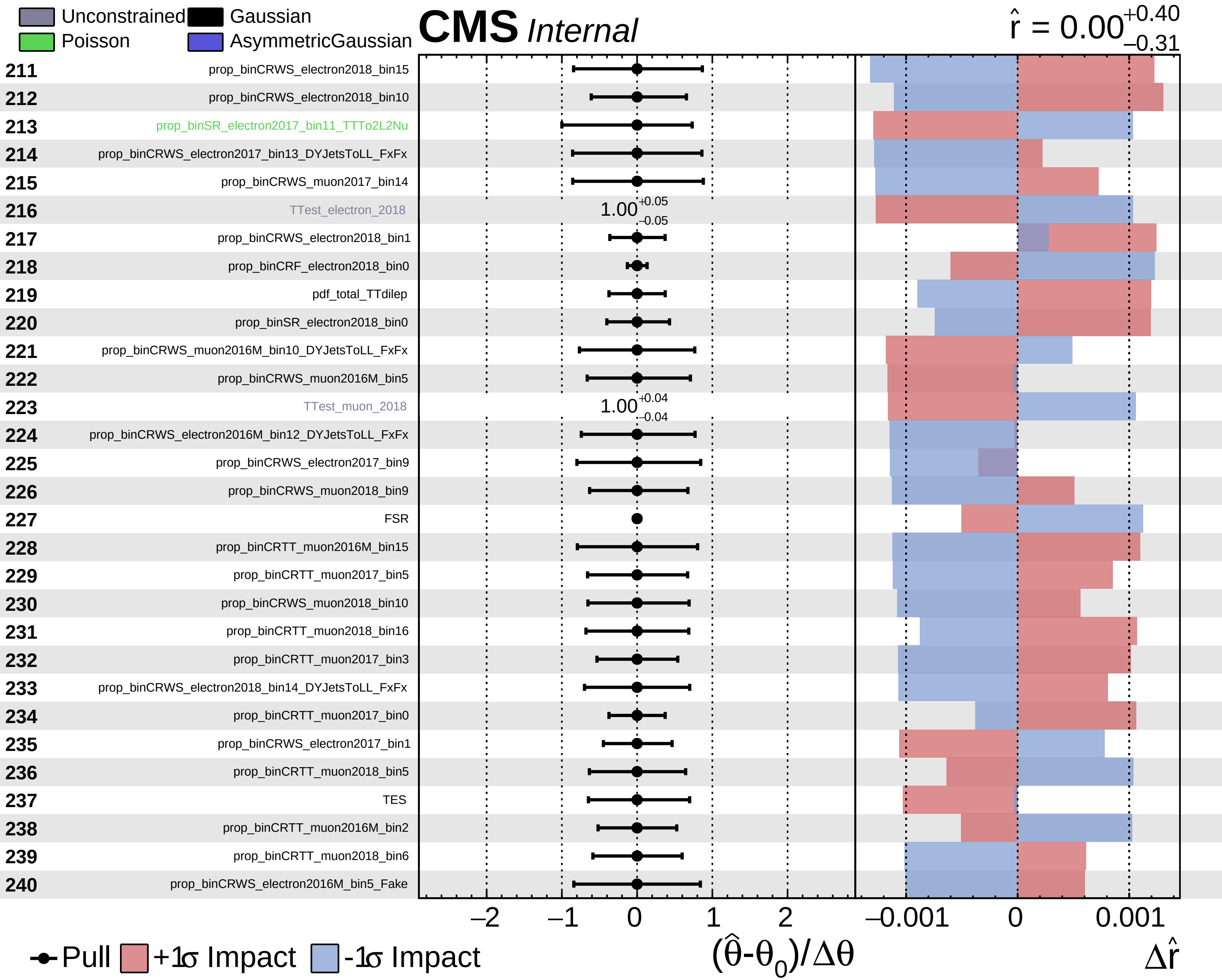
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
121 PF
122 jes
123 prop_binSR_electron2016M_bin5
124 prop_binSR_electron2016M_bin4
125 prop_binCRWS_electron2017_bin0
126 prop_binSR_muon2017_bin12_WrongSign
127 prop_binCRTT_electron2018_bin0
128 pdf_total_DY
129 prop_binSR_muon2017_bin0
130 prop_binSR_electron2016M_bin2_Fake
131 prop_binCRTT_muon2018_bin0
132 prop_binCRWS_muon2018_bin4
133 prop_binSR_electron2016M_bin1
134 pdf_total_TVX
135 prop_binCRWS_electron2016M_bin2
136 prop_binSR_electron2017_bin11_ZZtoLep
137 prop_binCRTT_electron2018_bin2
138 prop_binSR_muon2017_bin12_TVX
139 pdf_total_Triboson
140 prop_binCRWS_muon2017_bin6
141 prop_binCRWS_electron2017_bin3
142 prop_binCRWS_muon2018_bin3
143 prop_binCRWS_muon2018_bin12_DYJetsToLL_FxFx
144 prop_binCRWS_muon2016M_bin1
145 prop_binCRWS_electron2017_bin5
146 prop_binCRWS_electron2016M_bin1
147 prop_binCRWS_electron2016M_bin10_DYJetsToLL_FxFx
148 prop_binCRWS_muon2018_bin1
149 prop_binSR_electron2016M_bin3_DYJetsToLL_FxFx
150 prop_binCRWS_muon2018_bin15
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



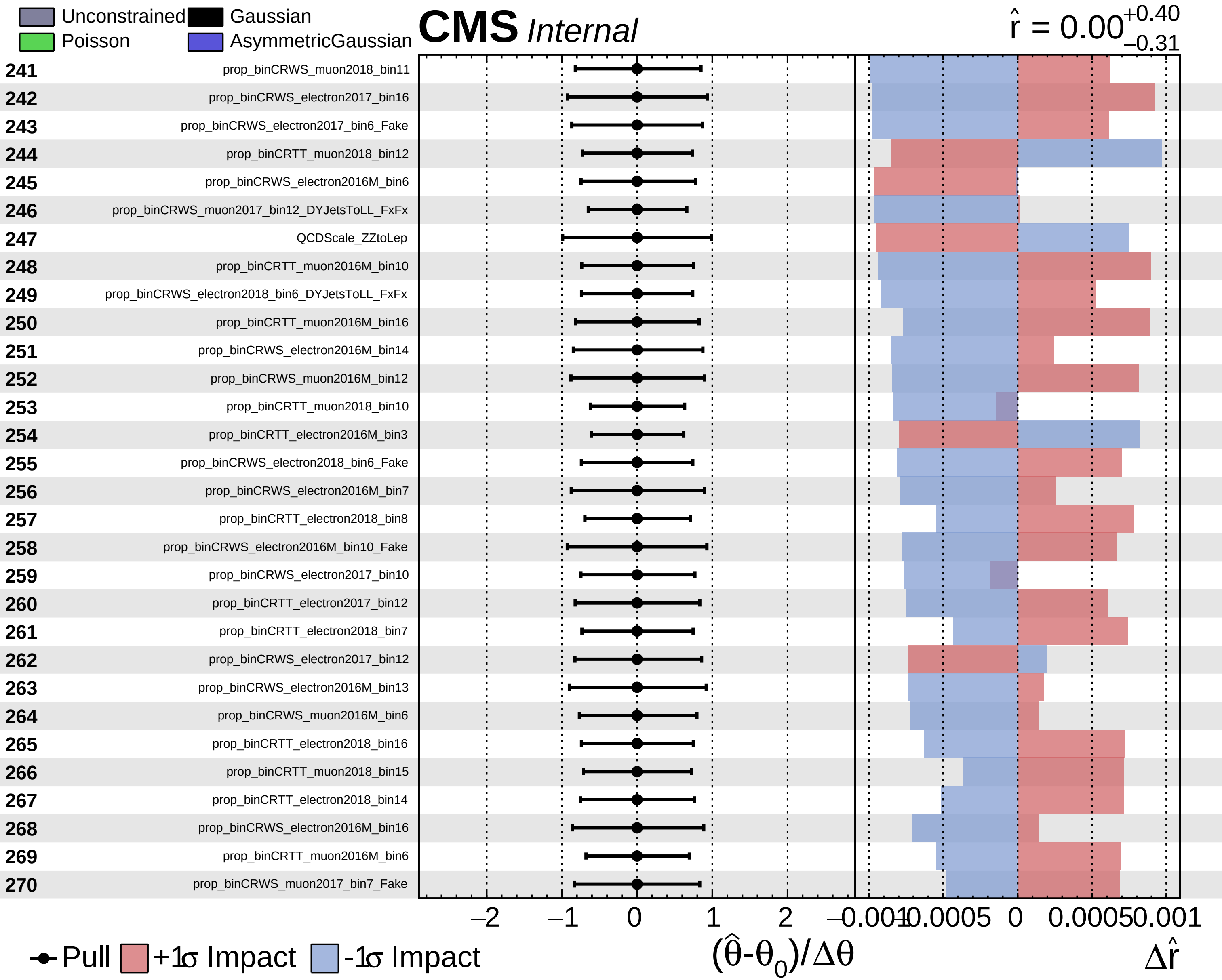
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
151 prop_binCRWS_electron2016M_bin3
152 jer_2017
153 prop_binCRTT_electron2016M_bin0
154 prop_binCRF_electron2017_bin0
155 prop_binSR_electron2016M_bin3_Fake
156 prop_binCRWS_electron2017_bin8
157 jer_2016M
158 prop_binCRWS_electron2017_bin2
159 QCDScale_TTdilep
160 prop_binSR_electron2017_bin1
161 prop_binCRWS_muon2016M_bin7
162 prop_binCRTT_electron2017_bin0
163 prop_binCRWS_electron2016M_bin5_DYJetsToLL_FxFx
164 prop_binCRWS_muon2017_bin8
165 prop_binCRWS_muon2018_bin2
166 prop_binCRTT_muon2018_bin3
167 prop_binCRTT_muon2018_bin1
168 prop_binCRF_electron2016M_bin0
169 tau_vsjet_2017
170 prop_binCRWS_muon2017_bin7_DYJetsToLL_FxFx
171 prop_binCRWS_muon2016M_bin3
172 prop_binCRWS_muon2018_bin16
173 prop_binSR_electron2016M_bin2_DYJetsToLL_FxFx
174 prop_binCRWS_electron2017_bin6_DYJetsToLL_FxFx
175 prop_binSR_electron2017_bin11_WrongSign
176 prop_binCRWS_muon2017_bin9
177 prop_binCRTT_electron2018_bin9
178 prop_binCRWS_muon2018_bin8
179 prop_binCRWS_electron2016M_bin11
180 prop_binCRWS_muon2018_bin14
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂



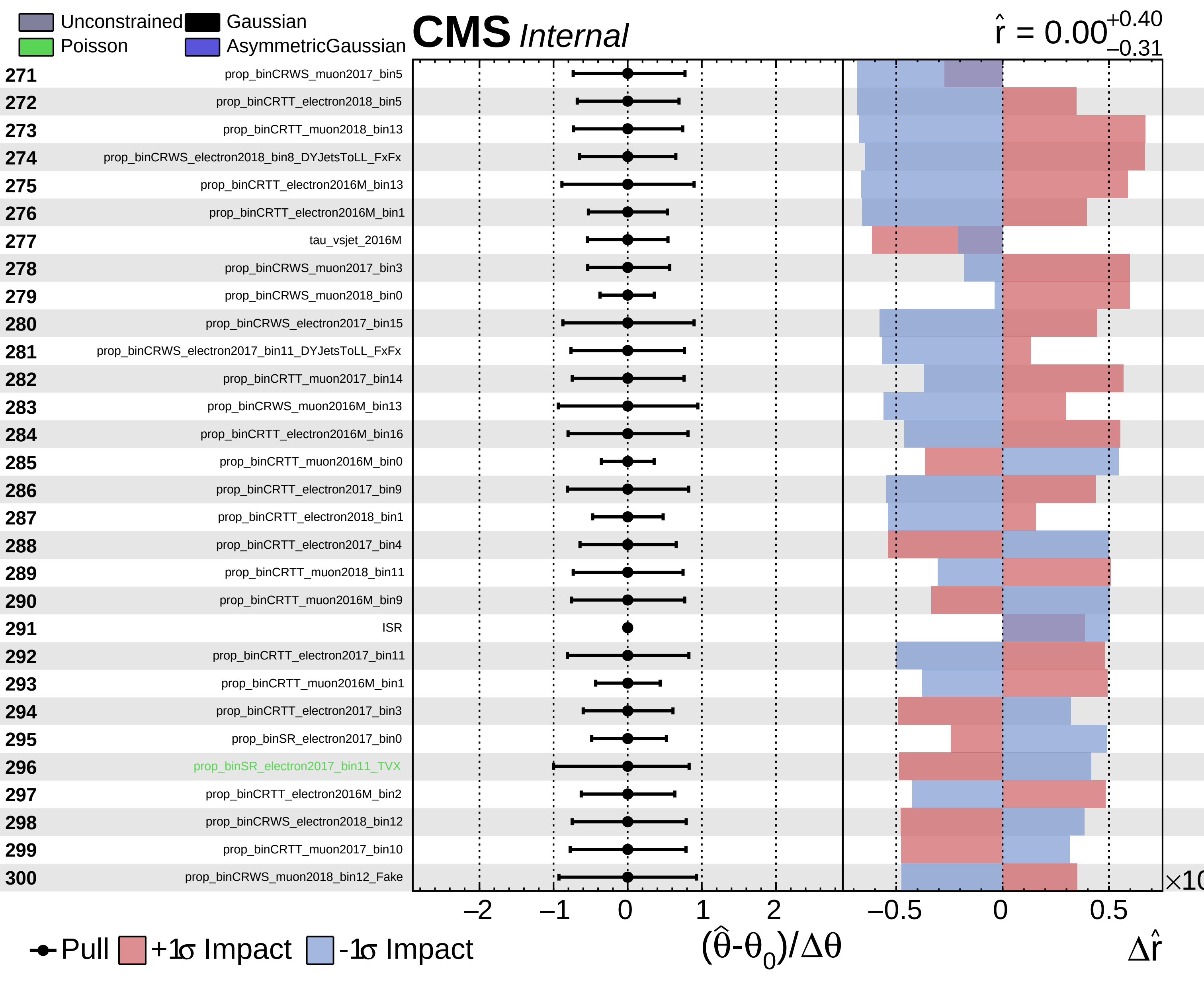
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
181 prop_binCRWS_muon2018_bin7
182 prop_binCRWS_electron2016M_bin0
183 prop_binCRWS_electron2016M_bin4
184 prop_binCRWS_muon2017_bin10
185 prop_binCRWS_muon2016M_bin0
186 TTest_electron_2017 $1.00^{+0.05}_{-0.05}$
187 prop_binCRTT_muon2017_bin1
188 mistag
189 prop_binSR_electron2018_bin1
190 prop_binCRTT_electron2017_bin2
191 prop_binCRWS_electron2016M_bin8
192 prop_binCRWS_muon2016M_bin8
193 prop_binCRTT_electron2017_bin1
194 prop_binCRWS_electron2018_bin16
195 prop_binSR_electron2016M_bin0
196 prop_binCRWS_electron2018_bin4
197 prop_binCRWS_muon2017_bin11
198 prop_binCRTT_electron2016M_bin6
199 TTest_muon_2016M $1.00^{+0.05}_{-0.05}$
200 prop_binCRWS_electron2017_bin7
201 prop_binSR_muon2018_bin0
202 TTest_muon_2017 $1.00^{+0.04}_{-0.04}$
203 pdf_total_WrongSign
204 prop_binCRWS_muon2016M_bin16
205 prop_binCRWS_muon2018_bin13
206 pu
207 prop_binCRTT_electron2018_bin3
208 prop_binCRWS_muon2016M_bin11
209 TTest_electron_2016M $1.00^{+0.06}_{-0.06}$
210 prop_binCRWS_electron2018_bin13
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

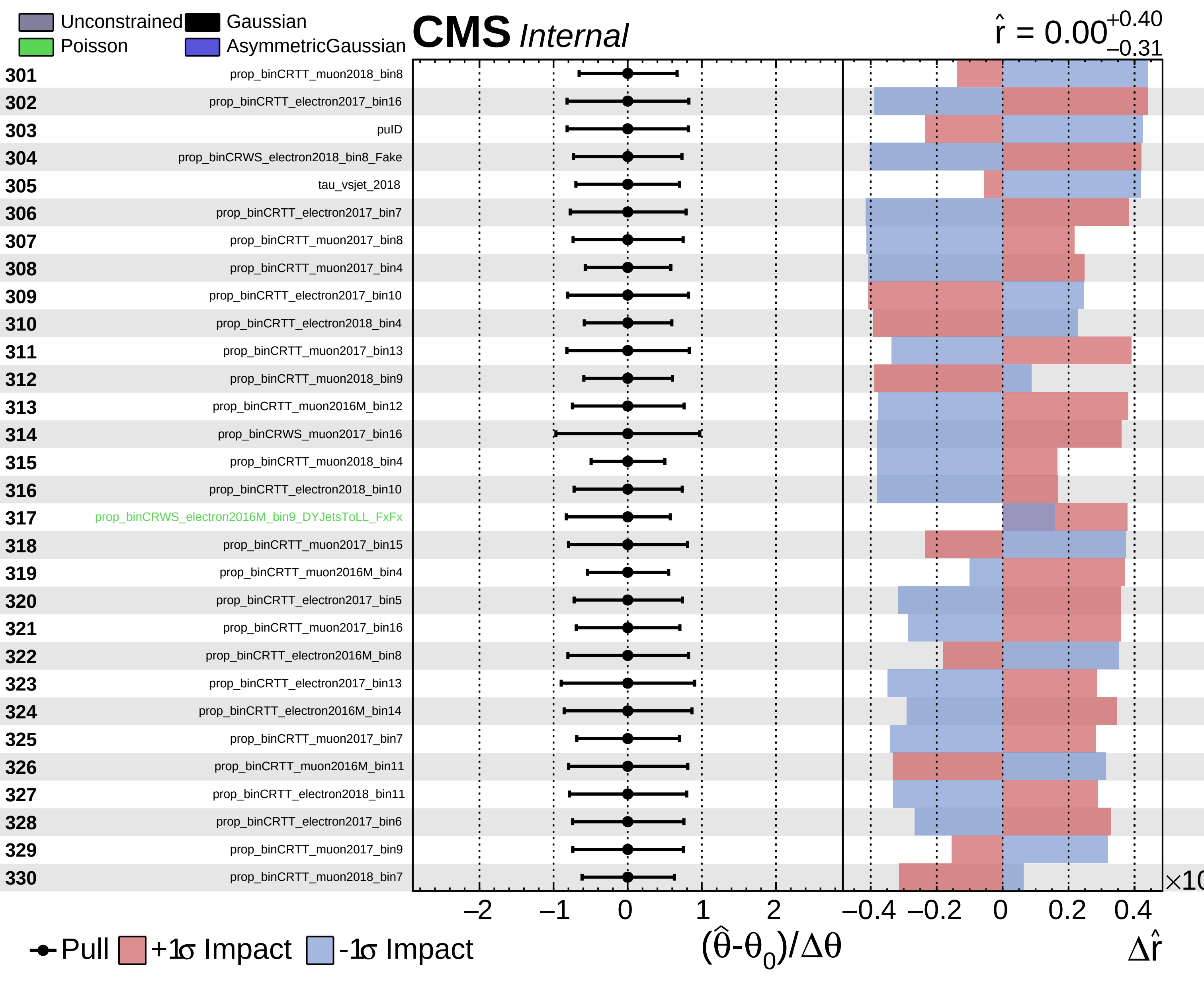


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
211 prop_binCRWS_electron2018_bin15
212 prop_binCRWS_electron2018_bin10
213 prop_binSR_electron2017_bin11_TTTo2L2Nu
214 prop_binCRWS_electron2017_bin13_DYJetsToLL_FxFx
215 prop_binCRWS_muon2017_bin14
216 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
217 prop_binCRWS_electron2018_bin1
218 prop_binCRF_electron2018_bin0
219 pdf_total_TTdilep
220 prop_binSR_electron2018_bin0
221 prop_binCRWS_muon2016M_bin10_DYJetsToLL_FxFx
222 prop_binCRWS_muon2016M_bin5
223 TTest_muon_2018
$1.00^{+0.04}_{-0.04}$
224 prop_binCRWS_electron2016M_bin12_DYJetsToLL_FxFx
225 prop_binCRWS_electron2017_bin9
226 prop_binCRWS_muon2018_bin9
227 FSR
228 prop_binCRTT_muon2016M_bin15
229 prop_binCRTT_muon2017_bin5
230 prop_binCRWS_muon2018_bin10
231 prop_binCRTT_muon2018_bin16
232 prop_binCRTT_muon2017_bin3
233 prop_binCRWS_electron2018_bin14_DYJetsToLL_FxFx
234 prop_binCRTT_muon2017_bin0
235 prop_binCRWS_electron2017_bin1
236 prop_binCRTT_muon2018_bin5
237 TES
238 prop_binCRTT_muon2016M_bin2
239 prop_binCRTT_muon2018_bin6
240 prop_binCRWS_electron2016M_bin5_Fake
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

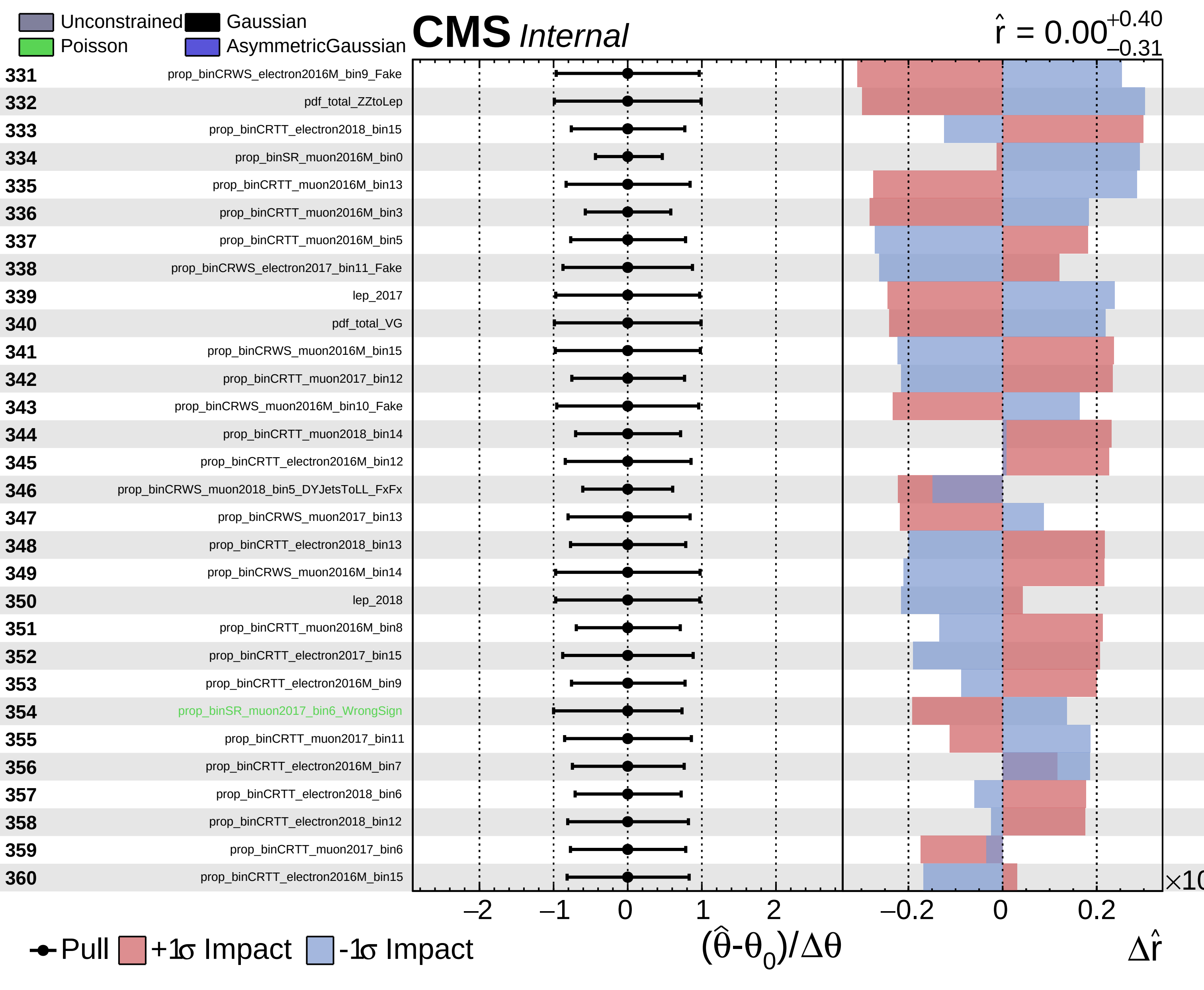


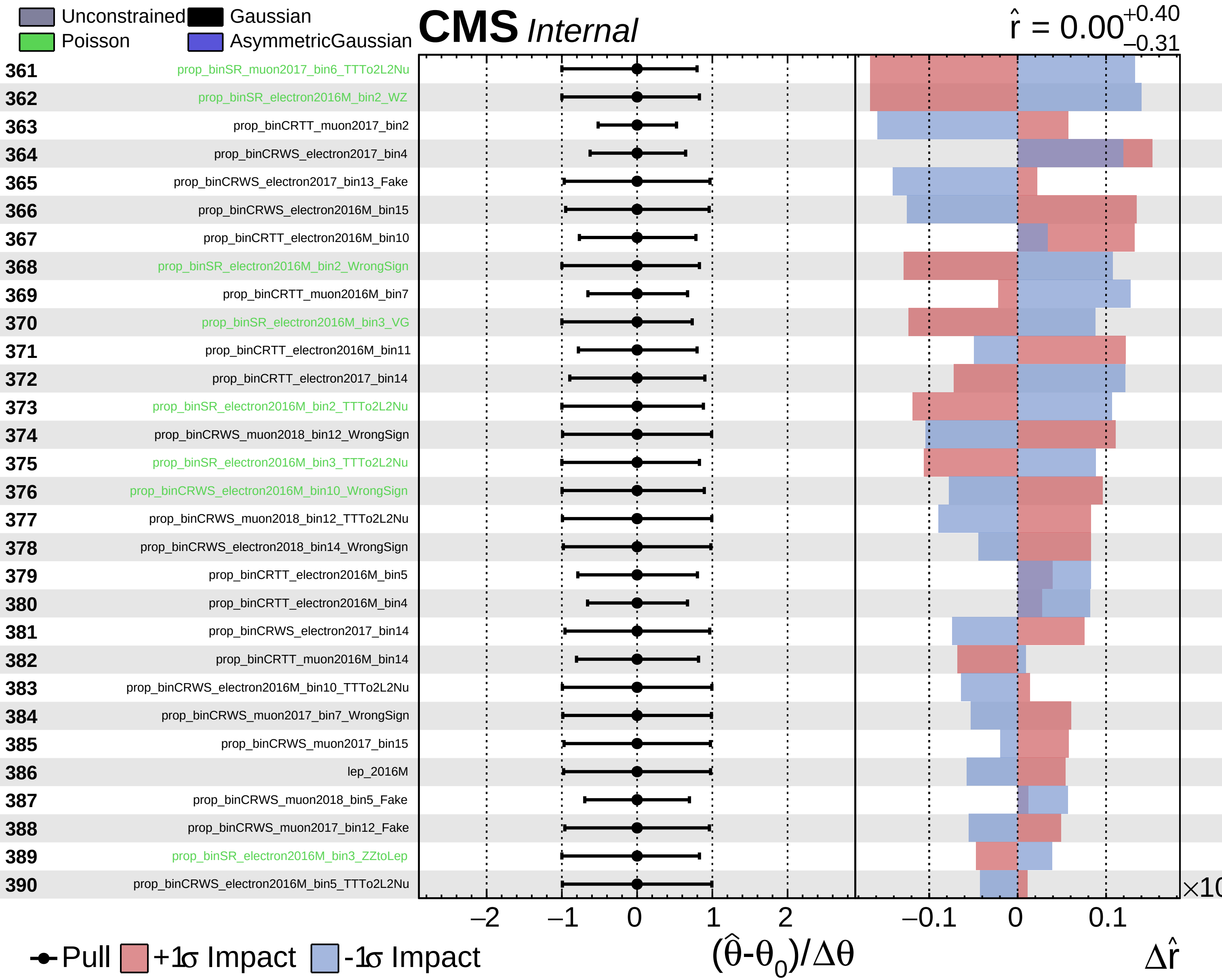
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
241 prop_binCRWS_muon2018_bin11
242 prop_binCRWS_electron2017_bin16
243 prop_binCRWS_electron2017_bin6_Fake
244 prop_binCRTT_muon2018_bin12
245 prop_binCRWS_electron2016M_bin6
246 prop_binCRWS_muon2017_bin12_DYJetsToLL_FxFx
247 QCDScale_ZZtoLep
248 prop_binCRTT_muon2016M_bin10
249 prop_binCRWS_electron2018_bin6_DYJetsToLL_FxFx
250 prop_binCRTT_muon2016M_bin16
251 prop_binCRWS_electron2016M_bin14
252 prop_binCRWS_muon2016M_bin12
253 prop_binCRTT_muon2018_bin10
254 prop_binCRTT_electron2016M_bin3
255 prop_binCRWS_electron2018_bin6_Fake
256 prop_binCRWS_electron2016M_bin7
257 prop_binCRTT_electron2018_bin8
258 prop_binCRWS_electron2016M_bin10_Fake
259 prop_binCRWS_electron2017_bin10
260 prop_binCRTT_electron2017_bin12
261 prop_binCRTT_electron2018_bin7
262 prop_binCRWS_electron2017_bin12
263 prop_binCRWS_electron2016M_bin13
264 prop_binCRWS_muon2016M_bin6
265 prop_binCRTT_electron2018_bin16
266 prop_binCRTT_muon2018_bin15
267 prop_binCRTT_electron2018_bin14
268 prop_binCRWS_electron2016M_bin16
269 prop_binCRTT_muon2016M_bin6
270 prop_binCRWS_muon2017_bin7_Fake
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
271 prop_binCRWS_muon2017_bin5
272 prop_binCRTT_electron2018_bin5
273 prop_binCRTT_muon2018_bin13
274 prop_binCRWS_electron2018_bin8_DYJetsToLL_FxFx
275 prop_binCRTT_electron2016M_bin13
276 prop_binCRTT_electron2016M_bin1
277 tau_vsjet_2016M
278 prop_binCRWS_muon2017_bin3
279 prop_binCRWS_muon2018_bin0
280 prop_binCRWS_electron2017_bin15
281 prop_binCRWS_electron2017_bin11_DYJetsToLL_FxFx
282 prop_binCRTT_muon2017_bin14
283 prop_binCRWS_muon2016M_bin13
284 prop_binCRTT_electron2016M_bin16
285 prop_binCRTT_muon2016M_bin0
286 prop_binCRTT_electron2017_bin9
287 prop_binCRTT_electron2018_bin1
288 prop_binCRTT_electron2017_bin4
289 prop_binCRTT_muon2018_bin11
290 prop_binCRTT_muon2016M_bin9
291 ISR
292 prop_binCRTT_electron2017_bin11
293 prop_binCRTT_muon2016M_bin1
294 prop_binCRTT_electron2017_bin3
295 prop_binSR_electron2017_bin0
296 prop_binSR_electron2017_bin11_TVX
297 prop_binCRTT_electron2016M_bin2
298 prop_binCRWS_electron2018_bin12
299 prop_binCRTT_muon2017_bin10
300 prop_binCRWS_muon2018_bin12_Fake
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

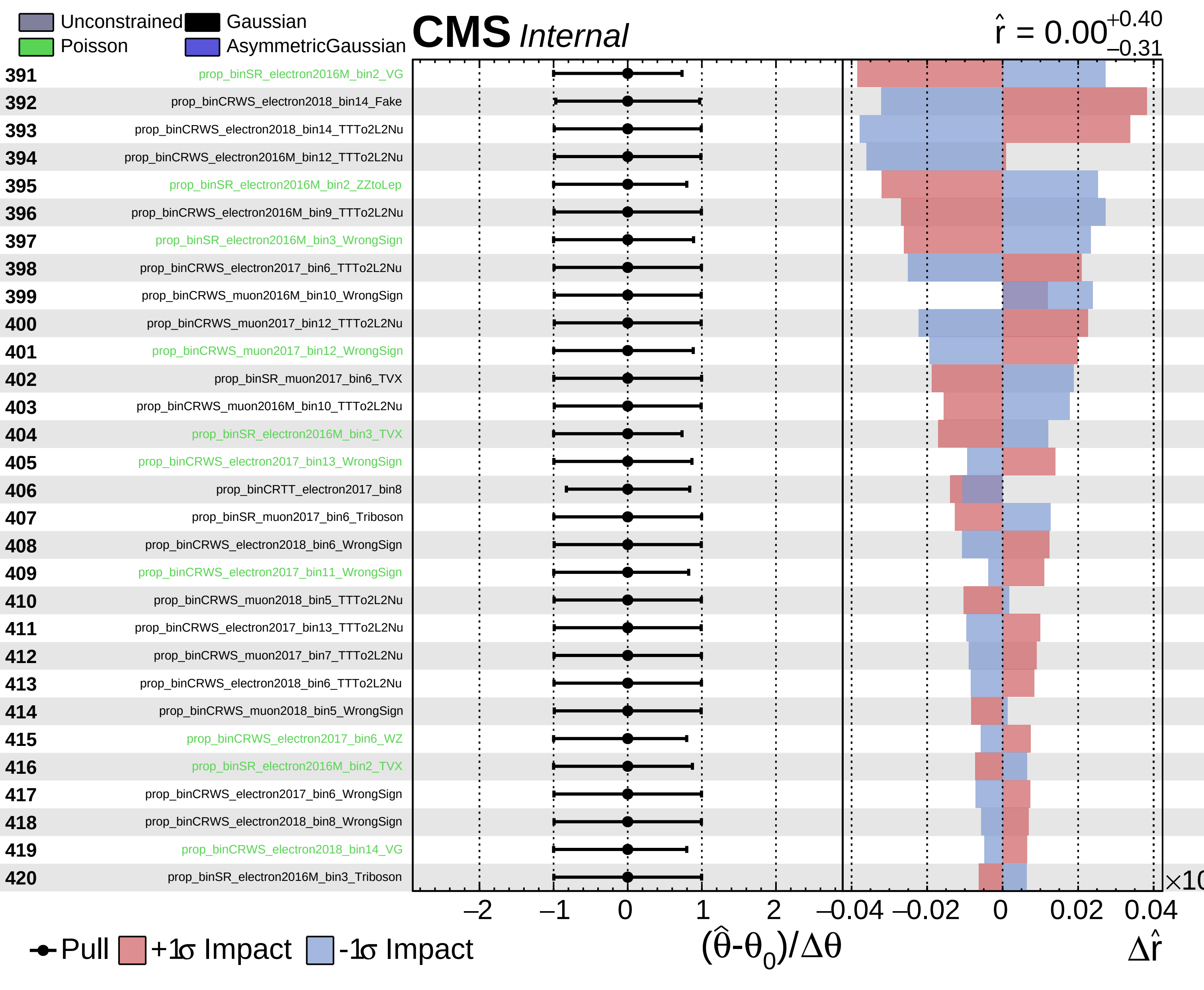


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
301 prop_binCRTT_muon2018_bin8
302 prop_binCRTT_electron2017_bin16
303 puID
304 prop_binCRWS_electron2018_bin8_Fake
305 tau_vsjet_2018
306 prop_binCRTT_electron2017_bin7
307 prop_binCRTT_muon2017_bin8
308 prop_binCRTT_muon2017_bin4
309 prop_binCRTT_electron2017_bin10
310 prop_binCRTT_electron2018_bin4
311 prop_binCRTT_muon2017_bin13
312 prop_binCRTT_muon2018_bin9
313 prop_binCRTT_muon2016M_bin12
314 prop_binCRWS_muon2017_bin16
315 prop_binCRTT_muon2018_bin4
316 prop_binCRTT_electron2018_bin10
317 prop_binCRWS_electron2016M_bin9_DYJetsToLL_FxFx
318 prop_binCRTT_muon2017_bin15
319 prop_binCRTT_muon2016M_bin4
320 prop_binCRTT_electron2017_bin5
321 prop_binCRTT_muon2017_bin16
322 prop_binCRTT_electron2016M_bin8
323 prop_binCRTT_electron2017_bin13
324 prop_binCRTT_electron2016M_bin14
325 prop_binCRTT_muon2017_bin7
326 prop_binCRTT_muon2016M_bin11
327 prop_binCRTT_electron2018_bin11
328 prop_binCRTT_electron2017_bin6
329 prop_binCRTT_muon2017_bin9
330 prop_binCRTT_muon2018_bin7
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

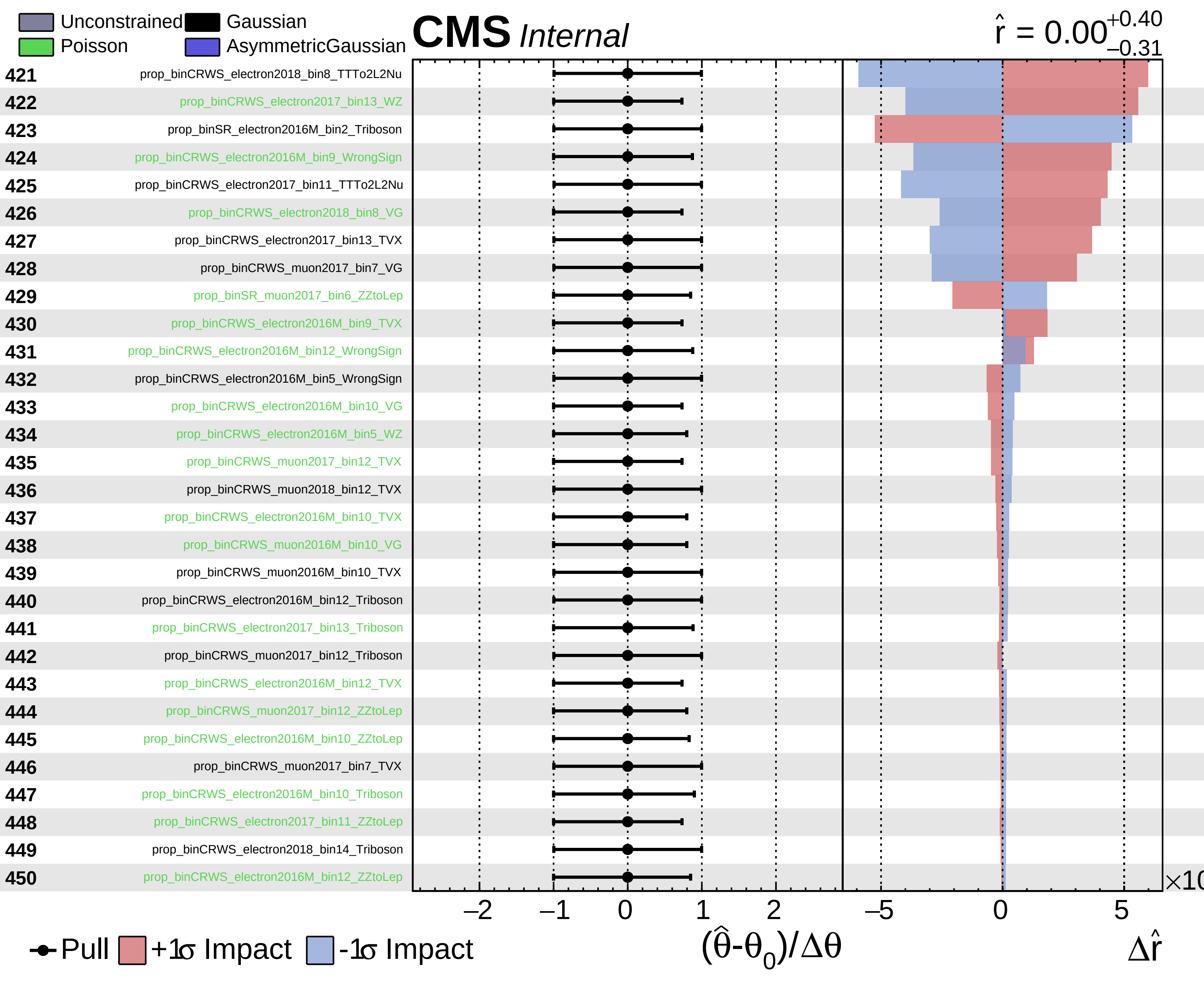

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
331 prop_binCRWS_electron2016M_bin9_Fake
332 pdf_total_ZZtoLep
333 prop_binCRTT_electron2018_bin15
334 prop_binSR_muon2016M_bin0
335 prop_binCRTT_muon2016M_bin13
336 prop_binCRTT_muon2016M_bin3
337 prop_binCRTT_muon2016M_bin5
338 prop_binCRWS_electron2017_bin11_Fake
339 lep_2017
340 pdf_total_VG
341 prop_binCRWS_muon2016M_bin15
342 prop_binCRTT_muon2017_bin12
343 prop_binCRWS_muon2016M_bin10_Fake
344 prop_binCRTT_muon2018_bin14
345 prop_binCRTT_electron2016M_bin12
346 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
347 prop_binCRWS_muon2017_bin13
348 prop_binCRTT_electron2018_bin13
349 prop_binCRWS_muon2016M_bin14
350 lep_2018
351 prop_binCRTT_muon2016M_bin8
352 prop_binCRTT_electron2017_bin15
353 prop_binCRTT_electron2016M_bin9
354 prop_binSR_muon2017_bin6_WrongSign
355 prop_binCRTT_muon2017_bin11
356 prop_binCRTT_electron2016M_bin7
357 prop_binCRTT_electron2018_bin6
358 prop_binCRTT_electron2018_bin12
359 prop_binCRTT_muon2017_bin6
360 prop_binCRTT_electron2016M_bin15
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



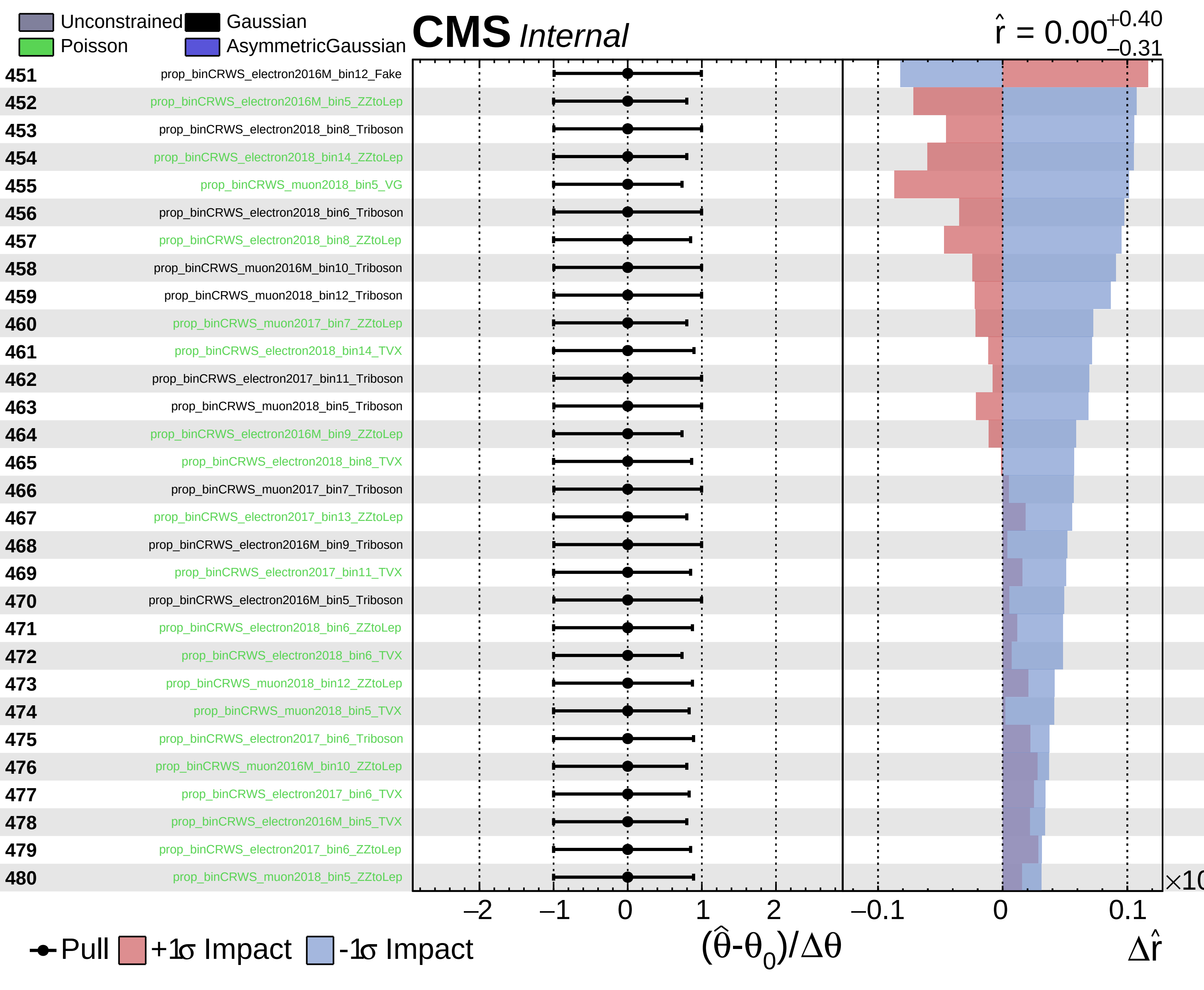
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
361 prop_binSR_muon2017_bin6_TTTo2L2Nu
362 prop_binSR_electron2016M_bin2_WZ
363 prop_binCRTT_muon2017_bin2
364 prop_binCRWS_electron2017_bin4
365 prop_binCRWS_electron2017_bin13_Fake
366 prop_binCRWS_electron2016M_bin15
367 prop_binCRTT_electron2016M_bin10
368 prop_binSR_electron2016M_bin2_WrongSign
369 prop_binCRTT_muon2016M_bin7
370 prop_binSR_electron2016M_bin3_VG
371 prop_binCRTT_electron2016M_bin11
372 prop_binCRTT_electron2017_bin14
373 prop_binSR_electron2016M_bin2_TTTo2L2Nu
374 prop_binCRWS_muon2018_bin12_WrongSign
375 prop_binSR_electron2016M_bin3_TTTo2L2Nu
376 prop_binCRWS_electron2016M_bin10_WrongSign
377 prop_binCRWS_muon2018_bin12_TTTo2L2Nu
378 prop_binCRWS_electron2018_bin14_WrongSign
379 prop_binCRTT_electron2016M_bin5
380 prop_binCRTT_electron2016M_bin4
381 prop_binCRWS_electron2017_bin14
382 prop_binCRTT_muon2016M_bin14
383 prop_binCRWS_electron2016M_bin10_TTTo2L2Nu
384 prop_binCRWS_muon2017_bin7_WrongSign
385 prop_binCRWS_muon2017_bin15
386 lep_2016M
387 prop_binCRWS_muon2018_bin5_Fake
388 prop_binCRWS_muon2017_bin12_Fake
389 prop_binSR_electron2016M_bin3_ZZtoLep
390 prop_binCRWS_electron2016M_bin5_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



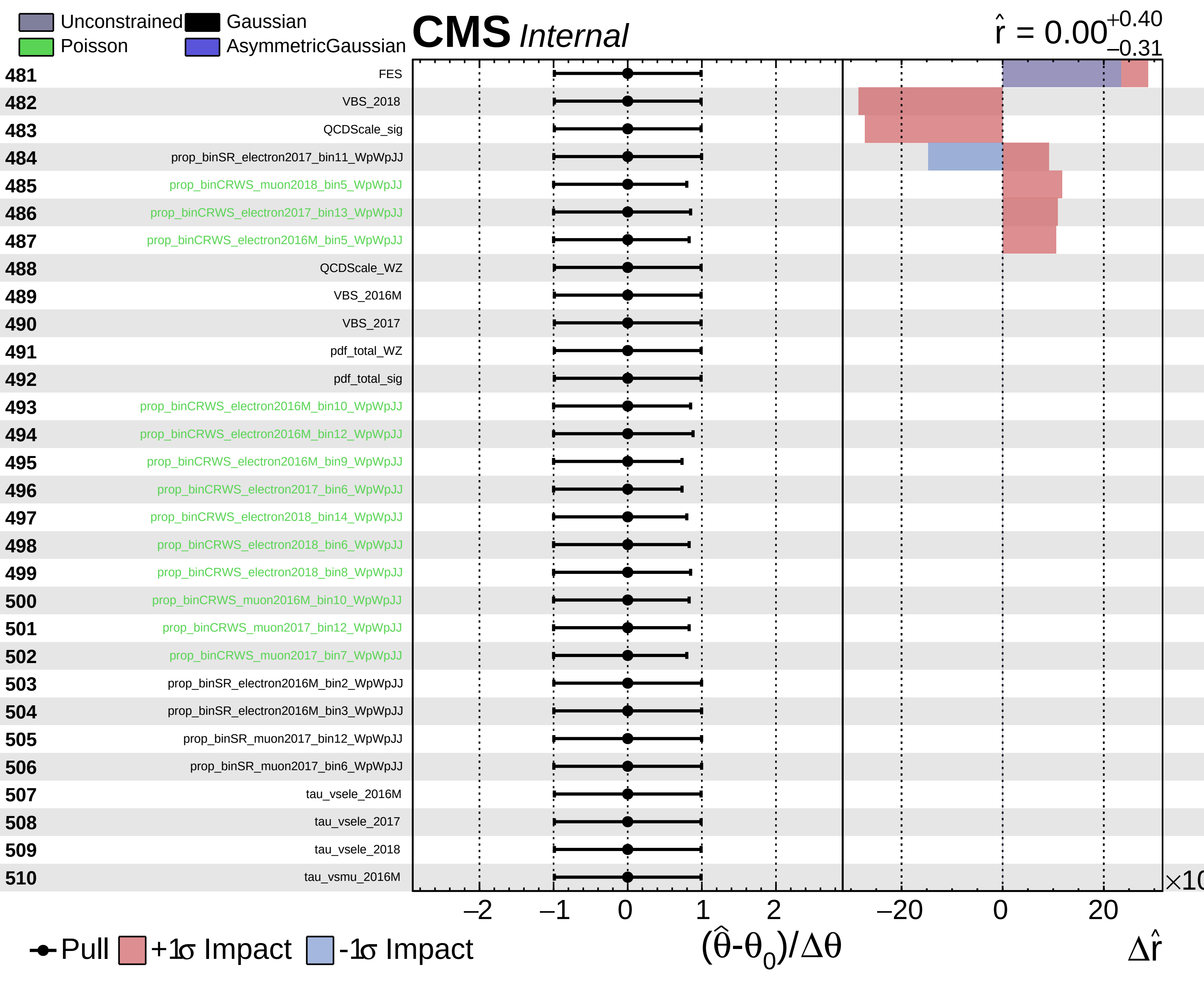
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
391 prop_binSR_electron2016M_bin2_VG
392 prop_binCRWS_electron2018_bin14_Fake
393 prop_binCRWS_electron2018_bin14_TTTo2L2Nu
394 prop_binCRWS_electron2016M_bin12_TTTo2L2Nu
395 prop_binSR_electron2016M_bin2_ZZtoLep
396 prop_binCRWS_electron2016M_bin9_TTTo2L2Nu
397 prop_binSR_electron2016M_bin3_WrongSign
398 prop_binCRWS_electron2017_bin6_TTTo2L2Nu
399 prop_binCRWS_muon2016M_bin10_WrongSign
400 prop_binCRWS_muon2017_bin12_TTTo2L2Nu
401 prop_binCRWS_muon2017_bin12_WrongSign
402 prop_binSR_muon2017_bin6_TVX
403 prop_binCRWS_muon2016M_bin10_TTTo2L2Nu
404 prop_binSR_electron2016M_bin3_TVX
405 prop_binCRWS_electron2017_bin13_WrongSign
406 prop_binCRTT_electron2017_bin8
407 prop_binSR_muon2017_bin6_Triboson
408 prop_binCRWS_electron2018_bin6_WrongSign
409 prop_binCRWS_electron2017_bin11_WrongSign
410 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
411 prop_binCRWS_electron2017_bin13_TTTo2L2Nu
412 prop_binCRWS_muon2017_bin7_TTTo2L2Nu
413 prop_binCRWS_electron2018_bin6_TTTo2L2Nu
414 prop_binCRWS_muon2018_bin5_WrongSign
415 prop_binCRWS_electron2017_bin6_WZ
416 prop_binSR_electron2016M_bin2_TVX
417 prop_binCRWS_electron2017_bin6_WrongSign
418 prop_binCRWS_electron2018_bin8_WrongSign
419 prop_binCRWS_electron2018_bin14_VG
420 prop_binSR_electron2016M_bin3_Triboson
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

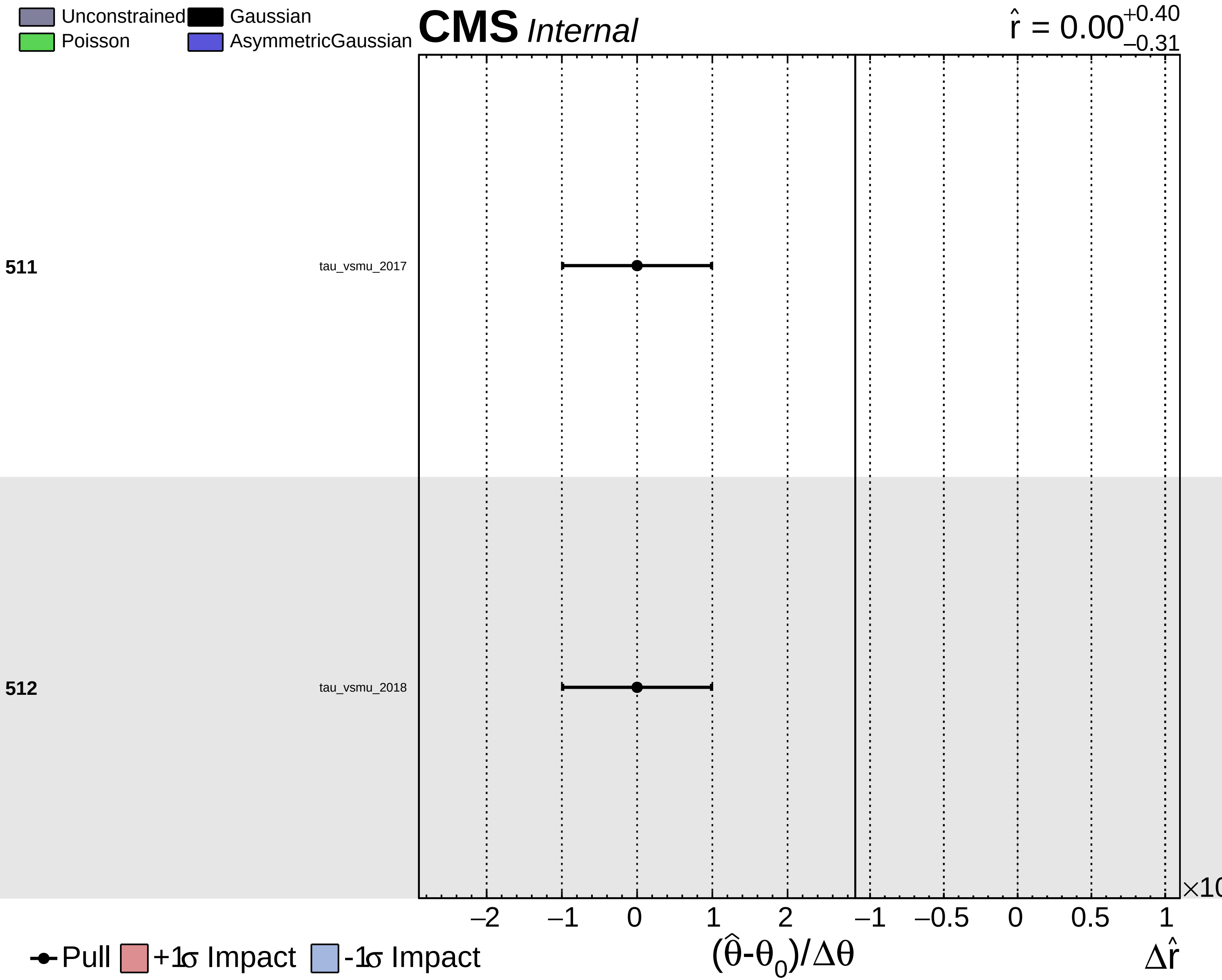


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
421 prop_binCRWS_electron2018_bin8_TTTo2L2Nu
422 prop_binCRWS_electron2017_bin13_WZ
423 prop_binSR_electron2016M_bin2_Triboson
424 prop_binCRWS_electron2016M_bin9_WrongSign
425 prop_binCRWS_electron2017_bin11_TTTo2L2Nu
426 prop_binCRWS_electron2018_bin8_VG
427 prop_binCRWS_electron2017_bin13_TVX
428 prop_binCRWS_muon2017_bin7_VG
429 prop_binSR_muon2017_bin6_ZZtoLep
430 prop_binCRWS_electron2016M_bin9_TVX
431 prop_binCRWS_electron2016M_bin12_WrongSign
432 prop_binCRWS_electron2016M_bin5_WrongSign
433 prop_binCRWS_electron2016M_bin10_VG
434 prop_binCRWS_electron2016M_bin5_WZ
435 prop_binCRWS_muon2017_bin12_TVX
436 prop_binCRWS_muon2018_bin12_TVX
437 prop_binCRWS_electron2016M_bin10_TVX
438 prop_binCRWS_muon2016M_bin10_VG
439 prop_binCRWS_muon2016M_bin10_TVX
440 prop_binCRWS_electron2016M_bin12_Triboson
441 prop_binCRWS_electron2017_bin13_Triboson
442 prop_binCRWS_muon2017_bin12_Triboson
443 prop_binCRWS_electron2016M_bin12_TVX
444 prop_binCRWS_muon2017_bin12_ZZtoLep
445 prop_binCRWS_electron2016M_bin10_ZZtoLep
446 prop_binCRWS_muon2017_bin7_TVX
447 prop_binCRWS_electron2016M_bin10_Triboson
448 prop_binCRWS_electron2017_bin11_ZZtoLep
449 prop_binCRWS_electron2018_bin14_Triboson
450 prop_binCRWS_electron2016M_bin12_ZZtoLep
−2 −1 0 1 2
−5 0 5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.40 −0.31
451 prop_binCRWS_electron2016M_bin12_Fake
452 prop_binCRWS_electron2016M_bin5_ZZtoLep
453 prop_binCRWS_electron2018_bin8_Triboson
454 prop_binCRWS_electron2018_bin14_ZZtoLep
455 prop_binCRWS_muon2018_bin5_VG
456 prop_binCRWS_electron2018_bin6_Triboson
457 prop_binCRWS_electron2018_bin8_ZZtoLep
458 prop_binCRWS_muon2016M_bin10_Triboson
459 prop_binCRWS_muon2018_bin12_Triboson
460 prop_binCRWS_muon2017_bin7_ZZtoLep
461 prop_binCRWS_electron2018_bin14_TVX
462 prop_binCRWS_electron2017_bin11_Triboson
463 prop_binCRWS_muon2018_bin5_Triboson
464 prop_binCRWS_electron2016M_bin9_ZZtoLep
465 prop_binCRWS_electron2018_bin8_TVX
466 prop_binCRWS_muon2017_bin7_Triboson
467 prop_binCRWS_electron2017_bin13_ZZtoLep
468 prop_binCRWS_electron2016M_bin9_Triboson
469 prop_binCRWS_electron2017_bin11_TVX
470 prop_binCRWS_electron2016M_bin5_Triboson
471 prop_binCRWS_electron2018_bin6_ZZtoLep
472 prop_binCRWS_electron2018_bin6_TVX
473 prop_binCRWS_muon2018_bin12_ZZtoLep
474 prop_binCRWS_muon2018_bin5_TVX
475 prop_binCRWS_electron2017_bin6_Triboson
476 prop_binCRWS_muon2016M_bin10_ZZtoLep
477 prop_binCRWS_electron2017_bin6_TVX
478 prop_binCRWS_electron2016M_bin5_TVX
479 prop_binCRWS_electron2017_bin6_ZZtoLep
480 prop_binCRWS_muon2018_bin5_ZZtoLep
×10
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
481 FES
482 VBS_2018
483 QCDScale_sig
484 prop_binSR_electron2017_bin11_WpWpJJ
485 prop_binCRWS_muon2018_bin5_WpWpJJ
486 prop_binCRWS_electron2017_bin13_WpWpJJ
487 prop_binCRWS_electron2016M_bin5_WpWpJJ
488 QCDScale_WZ
489 VBS_2016M
490 VBS_2017
491 pdf_total_WZ
492 pdf_total_sig
493 prop_binCRWS_electron2016M_bin10_WpWpJJ
494 prop_binCRWS_electron2016M_bin12_WpWpJJ
495 prop_binCRWS_electron2016M_bin9_WpWpJJ
496 prop_binCRWS_electron2017_bin6_WpWpJJ
497 prop_binCRWS_electron2018_bin14_WpWpJJ
498 prop_binCRWS_electron2018_bin6_WpWpJJ
499 prop_binCRWS_electron2018_bin8_WpWpJJ
500 prop_binCRWS_muon2016M_bin10_WpWpJJ
501 prop_binCRWS_muon2017_bin12_WpWpJJ
502 prop_binCRWS_muon2017_bin7_WpWpJJ
503 prop_binSR_electron2016M_bin2_WpWpJJ
504 prop_binSR_electron2016M_bin3_WpWpJJ
505 prop_binSR_muon2017_bin12_WpWpJJ
506 prop_binSR_muon2017_bin6_WpWpJJ
507 tau_vsele_2016M
508 tau_vsele_2017
509 tau_vsele_2018
510 tau_vsmu_2016M
−2
−1
0
1
2
−20
0
20
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.40}_{-0.31}$
511
tau_vsmu_2017
512
tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$